Yanze Wu

Postdoc Researcher at Northwestern University **Email**: wuyanze0@gmail.com

1.Education

Aug 2018-Aug 2023: Ph.D. (Physical Chemistry) University of Pennsylvania. Advisor: Joseph Subotnik
June 2018: B.S. (Chemistry, *Zhiyuan Honor Degree*) Shanghai Jiao Tong University (SJTU)

2.Research

Postdoc Researcher, Northwestern University (Sep 2023 – present). Advisor: George Schatz

- Nonadiabatic dynamics and wavepacket simulation of high energy molecular collisions
- Ultrafast electron relaxation in plasmonic nanoparticles and effects on photocatalysis

Ph.D in Theoretical Chemistry, University of Pennsylvania (June 2019 – Aug 2023). Advisor: Joseph Subotnik

- Chemical dynamics in the presence of spin-orbit coupling and Berry curvature effects;
- Mixed-quantum classical algorithm (surface hopping) development for nonadiabatic dynamics with complex-valued Hamiltonians;
- TDDFT-SOC code optimizations in QChem;

Undergraduate Researcher, SJTU (Mar 2016 – June 2018). Advisor: Huai Sun

- Reaction kinetic model building and mechanism simplification of ReaxFF MD simulation.
- Machine Learning Model of Alkane Thermodynamic Properties

Undergraduate Research Internship, Caltech (June 2017 – Sept 2017). Advisor: John F. Brady

- Phase Behavior of Activity-doped Colloidal Gel (Brownian dynamics)

3.Awards & Honors

2022: Hochstrasser Fellowship (Penn)
2021: Miller Fellowship (Penn)
2018: Zhiyuan Outstanding Graduate Award (SJTU)
2017: Kaiyuan Fellowship (SJTU)
2015, 2016: Zhiyuan Honor Scholarship (SJTU)

4.Teaching Experiences

Teaching Assistant of Chemical Principle (Fall 2018), Inorganic Chemistry Lab (Spring 2019), Graduate Quantum Mechanics I (Fall 2019)

5. Contributed Talks

Aug 2025: ACS Fall Meeting.
June 2022: 52nd Midwest Theoretical Chemistry Conference
May 2018: 31st Chinese Chemistry Society Meeting.

6. Publications

[23] Buchowiecki, M., **Wu, Y.,** & Schatz, G. C. Elastic cross sections, excitation exchange cross sections, and rate constants for excitation and quenching in N(4S)-N(2D) collisions (2025). *Atomic Data and Nuclear Data Tables,* 164, 101728.

[22] Adamovich, I. V., Wu, Y., & Schatz, G. C. (2025). Semiclassical analytic theory of multi-channel electronic energy transfer in nonadiabatic atomic collisions. *The Journal of Chemical Physics*, 163(22).

[21] Bian, X., **Wu, Y.**, Qiu, T., Tao, Z., & Subotnik, J. E. (2024). A semiclassical non-adiabatic phase-space approach to molecular translations and rotations: Surface hopping with electronic inertial effects. *The Journal of Chemical Physics*, *161*(23).

[20] **Wu, Y.**, Hochlaf, M., & Schatz, G. C. (2024). Modeling of Collision-Induced Excitation and Quenching of Atomic Nitrogen. *The Journal of Chemical Physics*, 161(1), 014104.

[19] Tao, Z., Bian, X., **Wu, Y.**, Rawlinson, J., Littlejohn, R. G., & Subotnik, J. E. (2024). Total angular momentum conservation in Ehrenfest dynamics with a truncated basis of adiabatic states. *The Journal of Chemical Physics*, 160(5).

[18] **Wu, Y.**, Rawlinson, J., Littlejohn, R. G., & Subotnik, J. E. (2024). Linear and angular momentum conservation in surface hopping methods. *The Journal of Chemical Physics*, 160(2).

[17] Bian, X., Tao, Z., **Wu, Y.**, Rawlinson, J., Littlejohn, R. G., & Subotnik, J. E. (2023). Total angular momentum conservation in ab initio Born-Oppenheimer molecular dynamics. Physical Review B, 108(22), L220304.

[16] Athavale, V., Bian, X., Tao, Z., **Wu, Y.**, Qiu, T., Rawlinson, J., LittleJohn, R. G., & Subotnik, J. E. (2023). Surface hopping, electron translation factors, electron rotation factors, momentum conservation, and size consistency. *The Journal of Chemical Physics*, *159*(11).

[15] Zhou, Z., **Wu, Y.**, Bian, X., & Subotnik, J. E. (2023). Nonadiabatic Dynamics in a Continuous Circularly Polarized Laser Field with Floquet Phase-Space Surface Hopping. *Journal of Chemical Theory and Computation*, *19*(3), 718-732.

[14] **Wu, Y.**, & Subotnik, J. E. (2023). A quantum-classical Liouville formalism in a preconditioned basis and its connection with phase-space surface hopping. *The Journal of Chemical Physics*, *158*(2).

[13] Chandran, S. S., **Wu, Y.**, & Subotnik, J. E. (2022). Effect of Duschinskii rotations on spin-dependent electron transfer dynamics. *The Journal of Physical Chemistry A*, *126*(50), 9535-9552.

[12] Bian, X., **Wu, Y.**, Rawlinson, J., Littlejohn, R. G., & Subotnik, J. E. (2022). Modeling Spin-Dependent Nonadiabatic Dynamics with Electronic Degeneracy: A Phase-Space Surface-Hopping Method. *The Journal of Physical Chemistry Letters*, *13*, 7398-7404.

[11] **Wu, Y.**, Bian, X., Rawlinson, J. I., Littlejohn, R. G., & Subotnik, J. E. (2022). A phase-space semiclassical approach for modeling nonadiabatic nuclear dynamics with electronic spin. *The Journal of Chemical Physics*, *157*(1), 011101.

[10] Chandran, S. S., **Wu, Y.**, Teh, H. H., Waldeck, D. H., & Subotnik, J. E. (2022). Electron transfer and spin–orbit coupling: Can nuclear motion lead to spin selective rates?. *The Journal of Chemical Physics*, *156*(17), 174113.

[9] Bian, X., **Wu, Y.**, Teh, H. H., & Subotnik, J. E. (2022). Incorporating Berry Force Effects into the Fewest Switches Surface-Hopping Algorithm: Intersystem Crossing and the Case of Electronic Degeneracy. *Journal of Chemical Theory and Computation*, *18*(4), 2075-2090.

[8] **Wu, Y.**, & Subotnik, J. E. (2021). Semiclassical description of nuclear dynamics moving through complex-valued single avoided crossings of two electronic states. *The Journal of Chemical Physics*, *154*(23), 234101.

[7] **Wu, Y.**, & Subotnik, J. E. (2021). Electronic spin separation induced by nuclear motion near conical intersections. *Nature Communications*, *12*(1), 700.

[6] **Wu, Y.**, Miao, G., & Subotnik, J. E. (2020). Chemical Reaction Rates for Systems with Spin–Orbit Coupling and an Odd Number of Electrons: Does Berry's Phase Lead to Meaningful Spin-Dependent Nuclear Dynamics for a Two State Crossing?. *The Journal of Physical Chemistry A*, *124*(37), 7355-7372.

[5] Bian, X., **Wu, Y.**, Teh, H. H., Zhou, Z., Chen, H. T., & Subotnik, J. E. (2021). Modeling nonadiabatic dynamics with degenerate electronic states, intersystem crossing, and spin separation: A key goal for chemical physics. *The Journal of Chemical Physics*, *154*(11), 110901.

[4] **Wu, Y**., Sun. H., Wu, L., & Joshua, D. (2019). Extracting the Mechanism and Kinetic Model of Complex Reactions from Atomistic Simulation Data. *J. Comp. Chem*, 40(16), 1586-1592.

[3] Gong, Z., **Wu, Y**., Wu, L. & Sun, H. (2018). Predicting Thermodynamic Properties of Alkanes by High-throughput Force Field Simulation and Machine Learning. *J. Chem. Info. Model,* 58(12), 2502-2516.

[2] Omar, A. K., **Wu, Y.**, Wang, ZG. & Brady, J. F. (2018). Swimming to Stability: Structural and Dynamical Control via Active Doping. *ACS Nano*, 13(1), 560-572.

[1] Cao, F., Gong, Z., **Wu, Y.** & Sun, H. (2017). A high-throughput computing procedure for predicting vapor-liquid equilibria of binary mixtures-Using carbon dioxide and n-alkanes as examples. *Fluid Phase Equilibria*, 452, 58-68.